pyomnigraph - Getting and hosting your own federated copies of subgraphs of Wikidata and other knowledge graphs*

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Abstract

Wikidata is an example of a very large, crowd-sourced general knowledge graph backed by a worldwide community since 2012. FactGrid has been using the same Wikibase infrastructure as Wikidata since 2017 for historical research projects. GOV (Geschichtliches/Genealogisches Ortsverzeichnis) is an online gazetteer for searching ancient places. Wikidata grew so large that the Blazegraph endpoint powering it was pushed to the 4 terabyte limit and the Wikimedia Foundation therefore decided to split the knowledge graph to host two instances. The challenges of federated queries that came up in the above use cases led to the need to be able to work with subgraphs of the RDF graphs and try out different alternative endpoints such as QLever, Virtuoso and others. While typical research projects that do such subgraphing and benchmarks roll their own environments, our pyomnigraph, a Python-based open source library, provides a unified infrastructure with a standard set of REST APIs and CLI commands to improve the handling and potentially grow the matrix of subgraphs versus endpoints to work with. In this work we report on pyomnigraph usage on seven graphs, four endpoints and three use cases with four example queries each, spanning three different graphs per use case. This opens a multidimensional space for exploring different combinations of federated queries, starting from a natural remote-remote-remote federated query to the possible combinations of local federated queries backed by all endpoints supported by pyomnigraph. We show that typical frustrations of scaling multi-hop federated queries can be avoided if the subgraphs are locally loaded and federation is applied locally.

Keywords

Wikidata, FactGrid, QLever, Blazegraph, Apache Jena, Oxigraph, GOV, RDF triple store, federated SPARQL, benchmark, PyOmniGraph

1. Introduction

Wikidata [1] started in 2012 as an effort to back the different multilingual Wikipedia sites with a common set of facts. Initially, about 2 million entities were harvested from existing Wikipedia content. As of 2026, over 100 million entities have been curated by a growing worldwide community of contributors. In 2022, we reported on *Getting and Hosting Your Own Copy of Wikidata* [2]. For quite a few use cases, the need for federated queries based on subgraphs of diverse public and internal knowledge graphs has been our motivation to try out different combinations of endpoints and subgraphs. In the case of Wikidata we have even been forced to do so by the 2025 graph split [3].

Public alternatives to the official Wikidata Query Service WDQS are still rare. Our attempts to convince the Wikimedia Foundation to support a public mirror infrastructure have so far been in vain. Table 1 lists the seven working public Wikidata endpoints currently configured in nicescholia[1] [4] to be watched as potential alternative Scholia backends.

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[1]https://nicescholia.wikidata.dbis.rwth-aachen.de/api/endpoints

Table 1
Public Wikidata query endpoints. Triple counts for QLever (Freiburg) and Virtuoso measured with `COUNT(*)` on 2026-07-31; the DBIS QLever count query did not return within the timeout (–).

Provider	Triple Store	Triples (B)	Last update
Wikimedia Foundation (legacy full)	Blazegraph	22.0	continuous
Wikimedia Foundation (main)	Blazegraph	8.6	continuous
Wikimedia Foundation (scholarly)	Blazegraph	8.7	continuous
RWTH Aachen DBIS	Blazegraph	17.0	2025-06-01
RWTH Aachen DBIS	QLever	17.7	continuous
University of Freiburg	QLever	17.7	continuous
OpenLink Software (truthy)	Virtuoso	8.1	2025-11-01

The Wikimedia Foundation itself now serves three: the legacy full graph[2] and, since the WDQS graph split, a main[3] and a scholarly[4] half. RWTH Aachen University hosts a full Blazegraph mirror[5] alongside a QLever endpoint[6]; the University of Freiburg QLever[7] service has moved to the `qlever.dev` host. OpenLink Software[8] provides a Virtuoso instance restricted to truthy statements. These alternatives still have their own limitations regarding SPARQL language coverage, data currency and completeness.

Using federated queries on participants of the Linked Open Data Cloud LOD Cloud[9] is in theory an excellent way to answer queries over diverse data sources. In practice there are a lot of possible frustrations involved, such as outages of endpoints, incompatibilities of endpoints, awkward syntax and general query rot.

For real-life use cases that operate on small subgraphs only, it should be feasible to operate locally by providing the necessary data "on the fly". Unfortunately, to do so a lot of technical expertise is needed to set up endpoints, collect the subgraph data and finally perform the query. To ease this pain, we introduce pyomnigraph, which mitigates the complexity by using Docker and Python to provide a unified interface to different endpoint technologies in a local environment with limited resources.

The contributions of this work are:

- **pyomnigraph architecture**, a blueprint of components that separate concerns of data operations, server management and endpoint backends using Docker containers, see Figure 1.
- **pyomnigraph**, an Apache-2.0 licensed Python library that unifies the deployment and operation of multiple RDF triple stores behind a single API and Docker-based setup (Section 3), covering the backends listed in Section 3.1 and the `omnigraph` and `rdfdump` command-line interfaces (Section 3.2).
- **demo survey of working dataset and endpoint combinations**, see Tables 2 and 3.

2. Related Work

Wikidata graph split Willighagen et al. [3] report on the three options that have been considered for surviving the 2025 Wikidata graph split: federated queries between the main and the scholarly graph, using QLever [5] with modified queries on a full graph that does not suffer from the 4 TB limit of the Blazegraph endpoint, or using snapquery [6] with named parameterized queries as a middleware to separate the endpoints' concerns from the domain needs.

[2] https://query-legacy-full.wikidata.org/sparql
[3] https://query-main.wikidata.org/sparql
[4] https://query-scholarly.wikidata.org/sparql
[5] https://wdqs.wikidata.dbis.rwth-aachen.de/sparql
[6] https://qlever-api.wikidata.dbis.rwth-aachen.de/
[7] https://qlever.dev/api/wikidata
[8] https://truthy.demo.openlinksw.com/sparql
[9] http://cas.lod-cloud.net/

Wikidata subsetting Hosseini Beghaeiraveri et al. [7] survey approaches, tools and evaluation of Wikidata subsetting. Table 1 of the survey lists SPARQL CONSTRUCT queries as a separate tool category stating "Requires full dump" — a statement we do not agree with: access to a live SPARQL server is sufficient.

Linked Data Fragments The Linked Data Fragments line of work addresses the same pain point — public SPARQL endpoints cannot sustain highly concurrent live query workloads — with a different infrastructure approach: smart-KG [8] ships pre-materialized, compressed graph partitions from the server to the client for client-side evaluation, and Heling and Acosta [9] provide a framework for federated SPARQL query processing over heterogeneous Linked Data Fragments.

CONSTRUCT queries CONSTRUCT queries in SPARQL have been studied by Kostylev et al. [10]: the nearly identical syntax of CONSTRUCT and SELECT queries — both share the same WHERE clause — is "often deceptive". The issues to beware of are the output domain — partial mappings for SELECT versus RDF graphs for CONSTRUCT —, blank nodes in templates, which CONSTRUCT allows while they are unavailable in SELECT, and unbound variables that are silently dropped.

Federated SPARQL query processing Federated SPARQL query processing evaluates a single query over data that is distributed across several *remote SPARQL endpoints* [11]. What counts as a correct answer is defined by comparison with the centralized case. A federated query must return exactly the result that the same query would return over a single RDF graph holding the data of all participating endpoints [12, 13]. The difficulty is that this answer has to be produced without such a graph ever existing, and while contacting as few endpoints as possible. Two rather different mechanisms are commonly subsumed under the term, and they differ in who decides how the query is distributed. The first is part of the standard: SPARQL 1.1 provides the SERVICE keyword, which instructs a federated query processor to evaluate a designated part of a query against a designated remote endpoint and to combine the results it returns with those of the remaining parts [11]. Because the endpoint URLs are written into the query text, the author fixes the distribution in advance and the processor only dispatches what it is told. The second is a *federation engine*, a system layer built on top of the standard, which accepts an ordinary SPARQL query carrying no endpoint annotations, together with a catalogue of reachable endpoints, and derives the distribution itself. It does so in three stages (source selection and query decomposition, optimization, and execution), and engines differ mainly in how much knowledge the first stage draws on: zero-knowledge probing with ASK queries (FedX), precomputed indices or VoID summaries (SPLENDID, HiBISCuS, CostFed), or runtime adaptivity (ANAPSID) [13, 14, 15]. Throughout, the decisive optimization is the detection of *exclusive groups*, subqueries answerable by a single endpoint, which collapse many requests into one. The sources of a federation need not even be full SPARQL endpoints: interfaces of more restricted expressivity, such as triple pattern fragments, give rise to *heterogeneous* federations [13]. It is the automatic derivation of the distribution, rather than the delegation mechanism itself, that the results discussed below call into question.

Benchmarks Performance is assessed with benchmarks: FedBench has long been the reference [16], LargeRDFBench scales it to a billion triples with complex and large-data queries [15], FedShop scales the number of participating endpoints from 20 to 200 [17], QFed generates query sets dynamically for specialized systems [18], FQCEB targets cardinality estimation [19], and FKGQA extends the setting to federated question answering [20]. FedBench alone, however, cannot expose the problem at issue: all of its queries decompose into at most three subqueries, so every engine behaves alike on it [14], and with a mean of 4.3 triple patterns and runtimes of roughly two seconds its results do not extrapolate to the Data Web [15].

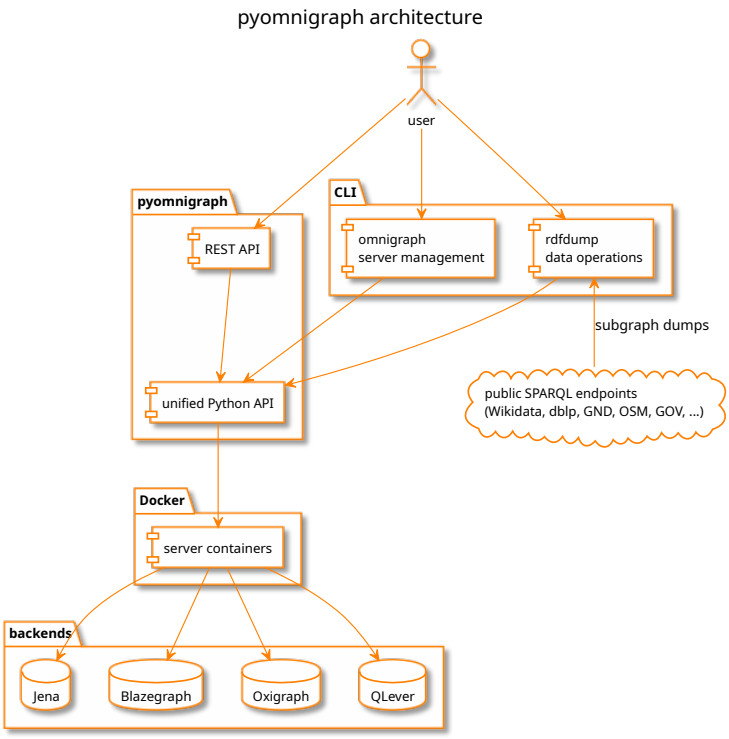

Figure 1: pyomnigraph architecture: CLI and planned REST API over the unified Python API, Docker managed triple store backends, subgraph dumps from public SPARQL endpoints. Source: `figures/architecture.puml` (PlantUML).

Alternative tooling The obvious alternative to pyomnigraph is the hand-rolled setup: benchmark and evaluation projects routinely build their own Docker compose files, store-specific loaders and engine control scripts, and the prevailing answer to the question of the best tooling tends to be "my own". Even the triple store authors themselves maintain per-engine control tooling rather than a cross-engine toolchain. pyomnigraph replaces this per-project fragmentation with one declarative subgraphs-versus-backends matrix; the direct feedback we received from triple store authors on this approach is tracked in our repositories.

3. The pyomnigraph Resource

pyomnigraph [21] provides a unified Python interface for multiple graph databases. The software is open source under Apache 2.0 license and hosted on GitHub. Figure 1 shows the architecture in one panel: the `omnigraph` and `rdfdump` command line interfaces and the planned REST API (future work, see Section 8) share the unified Python API, which manages the triple store backends as Docker containers; `rdfdump` pulls subgraph dumps from public SPARQL endpoints.

pyomnigraph leverages the named parameterized query approach from the snapquery [6] framework, which handles such parameterized SPARQL queries. It supports semantic annotation of queries and monitoring of their execution. By abstracting away the underlying endpoint, it can call multiple endpoints at once to compare results, or swap endpoints transparently, without users noticing or having to change anything.

The same separation of concerns—hiding details of endpoints and queries—is used by pyomnigraph. For human readability you will find YAML versions of these details in the `examples/<usecase>/` folders of our paper repository.

Table 2
Supported triple store backends per the pyomnigraph README as of 2026-07-31.

Database	Status	Notes
Apache Jena	working	robust, standards compliant
Blazegraph	working	high performance, easy setup
Oxigraph	working	Rust-based, embedded, fast
QLever	working	extremely fast queries
GraphDB	disabled	license required (free version available)
MillenniumDB	disabled	import-first workflow, Property Graph + RDF
Virtuoso	disabled	permissions issue
Stardog	planned	license required

3.1. Supported Backends

Table 2 reproduces the supported triple stores table of the pyomnigraph README[10].

3.2. Command-Line Interfaces

There are two command line interfaces available for pyomnigraph: `omnigraph` for server management and `rdfdump` for data operations.

The two core options are:

Listing 1: omnigraph core options

```
--cmd CMD [CMD ...]   commands to execute on servers: bash, clear,
                      count, info, load, logs, needed, rm, start,
                      status, stop, upload, webui

-s SERVERS [SERVERS ...], --servers SERVERS [SERVERS ...]
                      servers to work with - 'all' selects all
                      configured servers [default: ['blazegraph']]
```

so that multiple commands can be combined to be run on multiple servers.

The core options of `rdfdump` are:

Listing 2: rdfdump core options

```
-ds DATASETS [DATASETS ...], --datasets DATASETS [DATASETS ...]
                      datasets to work with - all is an alias for
                      all datasets [default: ['wikidata_triplestores']]

--dump                perform the dump [default: False]

-4o, --for-omnigraph  store dump at default omnigraph location
                      [default: False]
```

The design of the two tools is to make them available as a single toolchain — see the loop usage in Listing 4.

4. Use Cases and Data Sources

Three use cases are the basis for our demonstration and evaluation: papers and authors (scholarly), historical locations for genealogy research and railway journey planning. For each use case we specify a list of information needs, from which we derive concrete example queries. The abstraction is then done by applying named parameterized queries.

[10]https://github.com/WolfgangFahl/pyomnigraph

Table 3
Public SPARQL endpoints serving the subgraph sources, probed 2026-07-31 with `SELECT * WHERE {?s ?p ?o} LIMIT 1`.

Dataset	Endpoint	Engine	Probe
dblp	qlever.dev/api/dblp	QLever	0.072 s
GND	sparql.dnb.de/api/gnd	QLever	0.153 s
Wikidata	query.wikidata.org/sparql	Blazegraph	0.355 s
FactGrid	database.factgrid.de/sparql	Blazegraph	0.100 s
OpenStreetMap	sophox.org/sparql	Blazegraph	0.129 s
OpenStreetMap	qlever.dev/api/osm-planet	QLever	0.619 s
ERA RINF	rinf.data.era.europa.eu/api/sparql	RDF4J	0.139 s
GOV	gov-sparql.genealogy.net/dataset/sparql	Jena	0.060 s

Table 3 shows the subgraph sources for the use cases.

Table 4 tracks the named parameterized queries that we use to demo and benchmark, one row per query and example binding. Each query is defined in `examples/<usecase>/queries.yaml` with its parameter list and default values, each endpoint in `examples/<usecase>/endpoints.yaml`, so a row is reproduced by a single `sparqlquery` [22] call; the `query.sh` script of each use case reruns all its queries and writes a `<QueryName>.md` result file per query.

The examples can be reproduced from the pyomnigraphPaper GitHub repository[11]. Result sets that exceed the page limit of this paper are imported into the repository's query issues, one issue per query; the TryQueries[12] page complements them with short try-it links that run each query interactively against its endpoint.

As typical for real-world use cases, the ideal persistent identifiers for linking have limited availability. ORCID and ROR would be such identifiers for the scholarly case; according to Jesper Zedlitz, the counterpart for GOV would be a reconciliation via the current municipalities in Germany. This should work well via the Amtlicher Gemeindeschlüssel (official municipality key, P439). PID candidates for railway stations are the UIC station code P722[13], standardized by the UIC in Paris, or the international station number IBNR P954[14], which could formerly be resolved via the Deutsche Bahn departure board formatter URL[15] — marked with "link rot" in Wikidata, which is grist to our mill. The Adif page for Irún[16], as used in our ebike wiki[17], uses yet another number.

Per the histogram section of the query workflow issue[18] every named parameterized query gets a Histogram companion that unbinds the main seed parameter and shows the distribution of record counts over all seed values — so the realistic count range behind the Records column of Table 4 is visible and the example binding can be judged as typical or outlier. Listing 3 shows the shape; other parameters stay at their defaults. For each of the 12 queries: does the histogram run on the primary endpoint, and what is the distribution (min/median/max, position of the example binding)? Table 5 answers with the verified runs of the histogram verification issue[19]. Timeouts/aggregation limits on the full graphs are expected for the big datasets and are a result too — documented, not hidden.

Listing 3: Histogram companion shape

```
SELECT ?count (COUNT(?seed) AS ?freq) WHERE {
  { SELECT ?seed (COUNT(*) AS ?count) WHERE {
      # same WHERE body as the named query, seed unbound
    } GROUP BY ?seed }
```

[11] https://github.com/WolfgangFahl/pyomnigraphPaper
[12] https://cr.bitplan.com/index.php/TryQueries
[13] https://www.wikidata.org/wiki/Property:P722
[14] https://www.wikidata.org/wiki/Property:P954
[15] https://reiseauskunft.bahn.de/bin/bhftafel.exe/en?input=$1&boardType=dep&time=actual&productsDefault=1111101&start=yes
[16] https://www.adif.es/en/w/11600-ir%C3%BAn
[17] https://ebike.bitplan.com/index.php/Irun
[18] https://github.com/WolfgangFahl/pyomnigraphPaper/issues/2
[19] https://github.com/WolfgangFahl/pyomnigraphPaper/issues/12

Table 4
Named parameterized queries per use case, with the example parameter binding used for the reproduction run and the resulting number of records. Run with `sparqlquery` against the endpoint declared in `examples/<usecase>/endpoints.yaml`.

Use case	Query	Dataset	Example	Records
scholarly	AuthorPapers	dblp	27/6858 (Fahl)	6
scholarly	CoauthorPapers	dblp	FahlHW0D22a	600
scholarly	EventAuthorPapers	Wikidata	Q133307039	235
scholarly	InstitutionAuthorPapers	GND	36225-6	27
locations	PlaceIdentity	GOV	AACHENJO30BS (Aachen)	6
locations	PlaceHierarchy	GOV	AACHENJO30BS (Aachen)	2
locations	PlaceNames	FactGrid	AACHENJO30BS (Aachen)	11
locations	PersonsOfPlace	Wikidata	AACHENJO30BS (Aachen)	5543
railway	RelationStops	OpenStreetMap	10492086 (MD 18061)	18
railway	LineIdentity	Wikidata	Q801991 (Bordeaux–Irun)	1
railway	StationsOfLine	ERA RINF	655000-1 (Bordeaux–Irun)	92
railway	StationLocality	OpenStreetMap	Q3095803 (Irun)	5

Table 5
Histogram companions of the named queries of Table 4: distribution of the record counts over all seed values on the primary endpoint, verified 2026-07-31 with `sparqlquery`. median: weighted median over seeds; %ile: share of seeds with a record count at or below the example binding. CoauthorPapers fails with the memory limit of the public dblp endpoint (HTTP 500 after trying to allocate 2.5 GB).

Query	seeds	min	median	max	example	%ile	runtime
AuthorPapers	4,157,438 authors	1	2	3,342	6	81.4	3.6 s
CoauthorPapers	–	–	–	–	600	–	fails
EventAuthorPapers	842 events	1	183	1,614,216	235	53.6	2.8 s
InstitutionAuthorPapers	6,292 institutions	1	1	2,546	27	95.9	1.6 s
PlaceIdentity	1,331,489 places	1	1	30	6	99.9	3 m 11 s
PlaceHierarchy	1,311,092 places	1	1	18	2	89.5	26.9 s
PlaceNames	2,069 places	1	1	42	11	98.7	4.8 s
PersonsOfPlace	21,960 places	1	2	165,624	5,543	99.8	8.8 s
RelationStops	3,158 relations	1	11	208	18	69.7	6.6 s
LineIdentity	19,156 lines	1	1	9	1	98.0	4.5 s
StationsOfLine	8,927 lines	1	5	1,024	92	99.1	6.2 s
StationLocality	45,283 stations	1	1	24	5	99.4	4.3 s

```
}
GROUP BY ?count
ORDER BY ?count
```

4.1. scholarly - authors and papers

Queries: the papers of an author, the papers of all coauthors of a given paper, the papers of all authors of a given scientific event, the papers of all authors of a given institution.

4.2. historical locations

For the historical location gazetteer GOV [23], a List of Queries[20] has been curated, which needs to be extended by references to FactGrid and Wikidata.

[20] https://gov-wiki.genealogy.net/index.php?title=List_of_Queries

4.3. railway journey planning

This use case is inspired by our velorail[21] work as presented at the Wikimedia Hackathon 2025 closing showcase [24].

5. Federated Query Demo and Benchmark

pyomnigraph is a useful tool for trying out federated queries in a local environment to avoid the typical frustrations to be expected from public endpoints. The *examples* directory of our GitHub paper repository shows how pyomnigraph's declarative approach and command line tools simplify the task.

Listing 4 shows the `load.sh` script that loads the railway use case datasets into all functional pyomnigraph backends.

Listing 4: Loading the railway use case datasets into all functional pyomnigraph backends

```
#!/usr/bin/env bash
# load the railway use case datasets into all functional pyomnigraph backends
# see https://github.com/WolfgangFahl/pyomnigraphPaper/issues/10
script_dir=$(dirname "$0")
for dataset in relation_stops_osm line_identity_wikidata line_relation_osm
    stations_of_line_rinf station_locality_osm
do
  rdfdump -dc "$script_dir/datasets.yaml" -ds $dataset --dump -4o
  omnigraph -dc "$script_dir/datasets.yaml" -ds $dataset -s all --cmd upload count
done
```

Note how calling the command line tools `rdfdump` and `omnigraph` in a loop creates a whole matrix of subgraphs versus backends. The `-s all` option covers the local backends marked working in Table 2 (Apache Jena, Blazegraph, Oxigraph, QLever); the Engine column of Table 3 refers to the public source endpoints the subgraphs are dumped from.

6. Evaluation

By applying the matrix approach of Section 5 we obtain interesting results on performance and failure cases. We then do a comparison of remote-remote-remote versus local-local-local federated queries across all possible combinations.

Federation works cleanly for the remote-remote-remote (RRR) and local-local-local (LLL) placements on three of the four backends, with a set of specific failure modes which we collected in the failure cases issue[22] of our paper repository. The $4^3 = 64$ off-diagonal grid is feasible for the executors Apache Jena, Oxigraph and QLever per the pair matrix — roughly a 10-minute run on a MacBook Pro 2.4 GHz 8-core with 32 GB of RAM; structural failures have been found with Blazegraph as reported in our GitHub repository.

Table 6 shows the LLL diagonal — all three SERVICE legs of each federated query on the same local backend — with the record counts of the `Count` companion queries and the query-only latency. Apache Jena, Oxigraph and QLever reproduce exactly the record counts of the RRR runs over the public origin endpoints (AuthorIdentity 1, PlaceIdentity 3, LineIdentity 1). Blazegraph answers PlaceIdentity with 81 instead of 3: it re-evaluates uncorrelated SERVICE legs per outer solution and multiplies their results — two-leg probes yield $9/9/1$ where $3/3/1$ is expected; AuthorIdentity and LineIdentity mask the defect because every leg returns a single row.

Table 7 shows which local executor reaches which local SERVICE target with a minimal one-row SERVICE probe: 14 of the 16 pairs work. Only Blazegraph as executor cannot call Oxigraph or QLever

[21] https://github.com/WolfgangFahl/velorail
[22] https://github.com/WolfgangFahl/pyomnigraphPaper/issues/16

Table 6
LLL diagonal: the three federated `Count` queries with all SERVICE legs on the executor backend, container-internal SERVICE URLs. Query-only latency via HTTP, median of three runs, measured 2026-07-31 on a MacBook Pro 2.4 GHz 8-core, 32 GB RAM; record count in parentheses, expected counts 1/3/1 per the RRR runs.

Backend	AuthorIdentity	PlaceIdentity	LineIdentity
Blazegraph	0.20 s (1)	0.24 s (81)	0.18 s (1)
Apache Jena	0.23 s (1)	0.45 s (3)	0.22 s (1)
Oxigraph	0.018 s (1)	0.015 s (3)	0.013 s (1)
QLever	0.079 s (1)	0.084 s (3)	0.125 s (1)

Table 7
Local executor versus SERVICE target compatibility, one-row SERVICE probe on 2026-07-31: + works, – fails (HTTP 400 No Content-Type given for the Blazegraph subrequests rejected by Oxigraph and QLever).

Executor \ Target	Blazegraph	Jena	Oxigraph	QLever
Blazegraph	+	+	–	–
Apache Jena	+	+	+	+
Oxigraph	+	+	+	+
QLever	+	+	+	+

legs — both reject its SERVICE subrequest with HTTP 400 *No Content-Type given* — while the more lenient Blazegraph and Jena targets accept it.

7. Availability

pyomnigraph is open source under the Apache 2.0 license, developed on GitHub[23], released on PyPI[24] and archived on Zenodo [21] under the concept DOI 10.5281/zenodo.21721772; version 0.2.3 is the release this paper refers to. The reproducible examples and the paper sources live in the pyomnigraphPaper repository[25]; the underlying libraries pyLoDStorage [22] and snapquery [6] are archived on Zenodo as well.

8. Conclusion and Future Work

As a resource paper we have been pragmatic about the subtle differences between CONSTRUCT and SELECT queries pointed out by Kostylev et al. [10]: the shared WHERE clause even led us to name the extraction pattern of our dataset declarations `select_pattern`. A formal analysis of the limitations of this pragmatic approach would be beneficial to avoid trial and error for use cases that are out of bounds of our approach.

The RESTful interface of the architecture (Figure 1) is future work: it is most interesting for agentic AI work, e.g. as an MCP service backend.

Declaration on Generative AI

AI assistance in this project is limited to spelling, grammar, and formatting/build tooling; AI is NOT used to generate scientific content, arguments, or prose.

[23] https://github.com/WolfgangFahl/pyomnigraph
[24] https://pypi.org/project/pyomnigraph/
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